

Na Min An

PH.D. STUDENT

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Research Interests

My research aims to develop assistive and trustworthy AI and machine learning systems grounded in an understanding of human perception, cognition, and action. I am particularly interested in identifying and mitigating modality-specific and social biases in vision-language models, using a mechanistic interpretability framework to uncover the internal representations and decision-making processes that give rise to biased behaviors.

Education

Korea Advanced Institute of Science and Technology (KAIST)

DOCTOR OF PHILOSOPHY IN ARTIFICIAL INTELLIGENCE, GPA: 4.13/4.3

- Advisors: Prof. Hyunjung Shim (2024-Current) and Dr. James Thorne (2023-2025)

[Seoul, South Korea](#)

Feb. 2028 (Expected)

University of Seoul

BACHELOR OF SCIENCE IN MATHEMATICS, GPA: 4.29/4.5

- Summa Cum Laude (Received academic excellence or Mayor's scholarship for every semester)

[Seoul, South Korea](#)

Feb. 2022

Skills

Language Python, MATLAB, R, C++, JavaScript
Certification Society of Actuaries (P, FM, and SRM)

Experience

KAIST

RESEARCH ASSISTANT

- Nominated for Apple PhD Fellowship program (3 students per institution)
- Ministry of Science and ICT Project: Contributed leadership to a national project aimed at developing socially unbiased generative models.
- Basic Research Lab Project: Co-led a national project on developing barrier-free AI systems in educational and mobility domains.
- Published six first-author and three co-authored Top-tier works, with research centered on fairness and human-centered ML systems.

[Seoul, South Korea](#)

March 2023 - PRESENT

University of Copenhagen

VISITING RESEARCHER

- Received a Danish Data Science Academy grant to advance research on explainability and reliability in multimodal models to build fairness-aware ML systems.
- Developed a neuron-level bias mitigation framework for VLMs and LVLMs with Dr. Yoonna Jang, in collaboration with the NVIDIA Taiwan team (Dr. Yusuke Hirota and Dr. Ryo Hachiuma).

[Copenhagen, Denmark](#)

March 2025 - May 2025

Korea Institute of Science and Technology (KIST)

RESEARCH INTERN

- Analyzed human perceptual limits and ML model performance under severely low-resolution visual conditions.
- Developed a reinforcement learning-based framework that emulates the learning effects frequently occurring in human psychophysical tests.
- Built an automated imaging pipeline for detecting fluorescence-image edges and identifying hyperpolarization phenomena with high precision.

[Seoul, South Korea](#)

Oct. 2022 - Feb. 2023

University of Seoul

RESEARCH INTERN

- Investigated district-level Covid-19 risk factors across Seoul and awarded 3rd place (out of 40+ teams) in a competitive big-data competition.
- Developed a hybrid CNN-RNN model to classify the progressive stages of pulmonary nodules from sequential CT scan slices.
- Analyzed YouTube channel performance to model subscriber trajectory and provided targeted strategies for subscriber acquisition.

[Seoul, South Korea](#)

Sep. 2020 - Dec. 2020

Selected Publications

- An, N. M.**, Jang Y., Hirota, Y., Hachiuma, R., Augenstein, I., and Shim, H., "Interpretable Debiasing of Vision-Language Models for Social Fairness," [CVPR](#), 2026.

2. Kim, E.*, Park, J.*, **An, N. M.***, Kim, J., Patel, H. L., Jin, J., Kruk, J., Agarwal, A., Panda, S., Ilasariya, F. A., Shim, H., and Oh, A., “World in a Frame: Understanding Culture Mixing as a New Challenge for Vision-Language Models,” *CVPR* (* denotes equal contribution), 2026.
3. Waheed, S.* and **An, N. M.***, “Image Embedding Sampling Method for Diverse Captioning,” *EMNLP Main* (* denotes equal contribution), 2025.
4. **An, N. M.***, Kim, E.*, Thorne, J., and Shim, H., “IOT: Embedding Standardization Method Towards Zero Modality Gap,” *ACL Outstanding* (Top 1%, 26 out of 3,000 accepted papers, * denotes equal contribution), 2025.
5. Bayramli, Z.*, Suleymanzade, A.*, **An, N. M.**, Ahmad, H., Kim, E., Kang, J., Thorne, J., and Oh, A., “Diffusion Models Through a Global Lens: Are They Culturally Inclusive?” *ACL Oral* (Top 8%, 243 out of 3,000 accepted papers, * denotes equal contribution), 2025.
6. Kang, W. J., Kim, E., **An, N. M.**, Kim, S., Choi, H., Kwak, K., and Thorne, J., “Sightation Counts: Leveraging Sighted User Feedback in Building a BLV-aligned Dataset of Diagram Descriptions,” *ACL Main*, 2025.
7. **An, N. M.**, Roh, H., Kim, S., Kim, J. H., and Im, M., “Machine Learning Techniques for Simulating Human Psychophysical Testing of Low-Resolution Phosphene Face Images in Artificial Vision,” *Advanced Science* (Impact Factor: 14.3, JCR Ranking: 6.5%), 2025 - Featured in *Korean National News*.
8. Jung, W., Hong, J., **An, N. M.**, Thorne, J., and Yun, S., “Stable Language Model Pre-training by Reducing Embedding Variability,” *EMNLP Main*, 2024.
9. Lee, N.*, **An, N. M.***, and Thorne, J. “Can Large Language Models Infer and Disagree Like Humans?” *EMNLP Main* (* denotes equal contribution), 2023.

Publications

1. Kim, E.*, Park, J.*, Park, N.*, **An, N. M.***, Kim, K., Yoon, S., Huo, J., and Shim, H., “Aligned but Stereotypical? The Hidden Influence of System Prompts on Social Bias in LVLM-Based Text-to-Image Models,” *ICLR RSI Workshop* (* denotes equal contribution), 2026.
2. Waheed, S., **An, N. M.**, Milford, M., Ramchurn, S. D., and Ehsan, S., “VLM-Guided Visual Place Recognition for Planet-Scale Geo-Localization,” *ACRA*, 2025.
3. Kim, J. S*, Yu, A*, Ismayilzada J*, Park, J., Kim, E., Ahmad, H., **An, N. M.**, Thorne, J., and Oh, A., “WHEN TOM EATS KIMCHI: Evaluating Cultural Awareness of Multimodal Large Language Models in Cultural Mixture Contexts,” *NAACL C3NLP Workshop Outstanding* (* denotes equal contribution), 2025.
4. **An, N. M.**, Weed, S., and Thorne, J., “Capturing the Relationship Between Sentence Triplets for LLM and Human-Generated Texts to Enhance Sentence Embeddings,” *EACL Findings*, 2024.
5. **An, N. M.**, Roh, H., Kim, S., Kim, J. H., and Im, M. “Reinforcement Learning Framework to Simulate Short-Term Learning Effects of Human Psychophysical Experiments Assessing the Quality of Artificial Vision,” *IJCNN Oral*, 2023.
6. **An, N. M.**, Roh. H., Jung, S., Kim, E. J., and Im, M. “Machine Learning Can Approaches as an Alternative to Human Psychophysical Tests of Prosthetic Vision,” *EMBS Abstract*, 2021.

Submitted Papers

1. **An, N. M.***, Kang, I.*, Lee, M., and Shim, H., “Blind to Position, Biased in Language: Probing Mid-Layer Representational Bias in Vision Language Encoders for Zero-Shot Language-Grounded Spatial Understanding,” *arXiv:2509.23098* (* denotes equal contribution), 2026.
2. Kim, E.*, **An, N. M.***, Kang, W. J., Kim, S., Thorne, J., and Shim, H., “Are Large Vision-Language Models Ready to Guide Blind and Low-Vision Individuals?,” *arXiv:2510.00766* (* denotes equal contribution), 2026.

Patents

1. Im, M., Roh, H., and **An, N. M.**, "Artificial Vision Parameter Learning and Automating Method for Improving Visual Prosthetic Systems", [US Patent Granted No. F06639](#), 2026.
2. Im, M., Roh, H., and **An, N. M.**, "Artificial Vision Expression Parameter Automation Learning System Method for Improving Artificial Vision Device", [Korea Paper Granted No. 0172619](#), 2021.

Service

Talks

- AI Tutorial Series at KAIST AI (Winter 2025).

Volunteer

- Provided on-site support for EMNLP conference (2023).
- Contributed to community support and educational outreach in Cambodia in Summer 2018.

Leader

- Lab representative student from March 2024 to Aug. 2025.
- Appeared in the KAIST AI promotional video (2024) and top-10 research achievements participant (2023).
- Lab research manager from Aug. 2023 to Dec. 2023.

Teaching Assistant

- Department Seminars & Colloquia, Fall 2023 & Spring 2026
- Computational Image Generation and Modification, Fall 2025

Mentor

- Led research mentorship for two Azerbaijan undergraduate students in South Korea from Sep. 2024 to Jan. 2025.
- Provided English and mathematics tutoring to Dongdaemun-gu (Seoul, South Korea) high school students in Fall 2019 and Winter 2020.

Reviewer

- IEEE Transactions on Audio, Speech and Language Processing 2025 and 2026
- ICLR RSI Workshop 2026
- ACL ARR February, May, July, October 2025, January 2026
- ACL Student Research Workshop 2025 and 2026
- IJCNN 2024, 2025, and 2026

References

Hyunjung Shim

ASSOCIATE PROFESSOR, KAIST AI

- Email: kateshim@kaist.ac.kr

[March 2024 - PRESENT](#)

Seoul, South Korea

James Thorne

CHIEF AI SCIENTIST, THEIA INSIGHTS (PAST: ASSISTANT PROFESSOR, KAIST AI)

- Email: james@jamesthorne.com

[March 2023 - Aug. 2025](#)

Cambridge, United Kingdom

Yoonna Jang

POST-DOC, UNIVERSITY OF COPENHAGEN

- Email: morelychee@gmail.com

[March. 2025 - Dec. 2025](#)

Copenhagen, Denmark

Isabelle Augenstein

PROFESSOR, DEPARTMENT OF COMPUTER SCIENCE, UNIVERSITY OF COPENHAGEN

- Email: augenstein@di.ku.dk

[March 2025 - Dec. 2025](#)

Copenhagen, Denmark

Alice Oh

PROFESSOR, KAIST SCHOOL OF COMPUTING

- Email: alice.oh@kaist.edu

[Sep. 2024 - Dec. 2025](#)

Daejeon, South Korea

Maesoon Im

PRINCIPAL RESEARCH SCIENTIST, BRAIN SCIENCE INSTITUTE, KIST

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[Oct. 2021 - Feb. 2023](#)

Seoul, South Korea